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1.Profile : Ph.D. in Electronics with experience in sensor-based systems, data analysis and Machine olfaction.

2. Education Qualification:

Academic Degree	University/Board/Institute	Year	Marks	Specialization
Ph.D.	Visvesvaraya National Institute of Technology (VNIT) Nagpur	2025	8.0	VLSI and Nanotechnology
M.Tech.	PDPM-Indian Institute of Information Technology, Design and Manufacturing Jabalpur	2017	7.1	ECE (Micro-Nano Electronics)
B.E.	Terna Public Charitable Trust's College of Engineering Osmanabad	2013	63.00%	Electronics and Telecommunications

3. Professional/ Research Experience

Name Of Organization	Date of Joining	Date of leaving	Designation	Nature of work
PDPM-Indian Institute of Information Technology,	01/06/2015	01/06/2017	Teaching Assistant (TA)	Master degree research

Design and Manufacturing Jabalpur				
Visvesvaraya National Institute of Technology (VNIT) Nagpur	13/07/2018	23/08/2024	Research Scholar	Research leading to doctoral
Indian Institute of Information Technology Nagpur	18/07/2024	24/06/2025	Junior Research Fellow (JRF)	Research and development work
Ramdevbaba University Nagpur	25/06/2025	Still working	Assistant professor	Teaching and Research

4. Project Work:

Ph.D.

Thesis title- Electronic Nose System: Focus on gas quantification & drift correction algorithms using Machine learning techniques. This work is carried out under the supervision of Dr. Sharvari Deshmukh and Dr. Raghavendra Deshmukh at VNIT Nagpur.

Project 1-In this work, MEMS based Electronic nose was developed for measurement and monitoring of prominent industrial pollutants and proposed a drift correction. A drift correction algorithm was developed using experimental design and robust regression to correct the drift induced in the sensors. The experiments consisted of eight experimental sets with mixtures of H₂S, CO, NH₃ and C₂H₅OH measured on the same sensor array at the start and after a gap of 1 year. The methodology consisted of mapping the response of sensors after 1 year compared with that observed at the very start (day 1) of the experiment.

Project 2-In this work, an Electronic nose in conjunction with gas analyzers is used to evaluate the odor performance of different parts of the city. Five different locations were selected from city. Further co-relation between the e-nose and gas analyzers was done with partial least-square

(PLS) regression. The analysis of variance result shows that the e-nose is able to predict the result of gas analyzer for NH₃, H₂S, CO, and CH₄.

M.Tech.

Thesis title-Design and Analysis of low power LNA with high linearity for wideband application. This work is carried out under the supervision of Dr. P.N. Kondekar at IIITDM Jabalpur.

In this research, a Low Noise Amplifier (LNA) is designed for the frequency range of 8.1-21GHz using CMOS 0.18μm technology. It is a two-stage amplifier which first stage consist of common gate topology and the second stage consist of cascode stage. The layout of LNA is done with DRC and LVS clean and also parasitic extraction is done for post layout simulation. The chip area of proposed LNA was 0.11mm² and it consumes power of 5.4mW from V_{dd}=1V.

5. Research Paper Published in International Journals:6

- 1) **T. Gawande**, R. Deshmukh and S. Deshmukh, "Analysis of Outdoor Air Quality Using Low-Cost MEMS-Based Electronic Nose and Gas Analyzer With Multivariate Statistical Approach," in IEEE Embedded Systems Letters, vol. 17, no. 1, pp. 18-21, Feb. 2025, doi: 10.1109/LES.2024.3431214.
- 2) **GAWANDE, TUSHAR PRAKASH** and Deshmukh, Raghavendra and Deshmukh, Sharvari," MEMS based Electronic-nose system for measurement of industrial gases with drift correction methodology", Measurement Science and Technology, <http://iopscience.iop.org/article/10.1088/1361-6501/ace073> (2023).
- 3) Pandey, S., **Gawande, T.**, Inge, S., Pathak, A. and Kondekar, P.N. (2018), Design and analysis of wideband low-power LNA for improved RF performance with compact chip area. IET Microw. Antennas Propag., 12: 1816-1820. <https://doi.org/10.1049/iet-map.2018.0055>.
- 4) Pandey, S., **Gawande, T.** & Kondekar, P.N. A 3.1–10.6 GHz UWB LNA Based on Self Cascode Technique for Improved Bandwidth and High Gain. Wireless Pers Commun 101, 1867–1882 (2018). <https://doi.org/10.1007/s11277-018-5795-1>.
- 5) Sunil Pandey, **Tushar Gawande**, Abhijeet Pathak & Pravin Neminath Kondekar (2017): A 0.9-V, 4.4-mW CMOS LNA with wideband input match and high gain for UWB applications, International Journal of Electronics Letters, DOI: 10.1080/21681724.2017.1378373
- 6) Pandey, S., **Gawande, T.** & Kondekar, P.N. A 0.9 V, 4.57 mW UWB LNA with Improved Gain and Low Power Consumption for 3.1–10.6 GHz Ultra-Wide Band Applications. Wireless Pers Commun 96, 583–597 (2017). <https://doi.org/10.1007/s11277-017-4185-4>

6. Research Paper Published in International Conferences:4

- 1) **T. Gawande**, P. Hazra and S. Deshmukh, "Multivariant Partial Least Square and Gradient Decent Regularization for transfer of training models in MEMS based gas sensor arrays," *2021 2nd International Conference for Emerging Technology (INCET)*, 2021, pp. 1-4, doi: 10.1109/INCET51464.2021.9456378.

2) **T. Gawande**, R. Deshmukh and S. Deshmukh, "Qualitative analysis of prominent industrial gases by MEMS based gas sensors," *2020 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, 2020, pp. 1-6, doi: 10.1109/CONECCT50063.2020.9198690.

3) **T. Gawande**, R. Deshmukh, R. Patrika and S. Deshmukh, "Inherent MEMS sensor array variability reduction using robust regression," *2019 IEEE Bombay Section Signature Conference (IBSSC)*, 2019, pp. 1-4, doi: 10.1109/IBSSC47189.2019.8973108.

4) S. Deshmukh and **T. Gawande**, "Design of Experiments for optimizing and selectivity study of mems sensor array based electronic nose for hazardous gases measurement," *2021 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, 2021, pp. 1-5, doi: 10.1109/CONECCT52877.2021.9622666.

7. Patents (Granted/Filled):

1) MEMS gas sensors based electronic-nose system for measurement of industrial hazardous gases. Patent Application No: 202121040133. **Date of Grant: 04/10/2024.**

2) A handheld electronic device for the measurement of industrial air pollutants Application No: 202221014907. Date of filing: 17/03/2022.

8. Consultancy Project

Project: **Odor evaluation and measurement for different types of toilets fitted in India Railways**-The objective of the project work was to assess the odor generated from the three different types of toilets and do a comparative analysis using the EN13725 norms and algorithms. Further also to evaluate the odor intensity of the collected samples for confirmation.

Client: Indian Railways, Southeast zone, Bilaspur.

Cost: 4 lakhs. (completed)

9. Research Project

Project: **Proposals on Methane Monitoring and Mitigation Technologies (M3T)**

The objective of this project is to develop a system for Methane Monitoring and Mitigation in wastewater treatment plant and Landfills (Research proposal is submitted to the Department of science and Technology [DST] for evaluation).

10. Scholarship/Awards/Prize:

Awarded "Visvesvaraya Ph.D. scheme", funded by the Miety government of India.

11. Subject Interest:

- I. CMOS circuit design
- II. Device fabrication technology
- III. Testing and diagnostic of digital circuits
- IV. Physics of semiconductor devices

12. Research Interested:

- I. To pursue further work in the direction of the development of a wearable electronic nose to determine the physiological state of a person.
- II. Development of new machine learning algorithms based on neural network, fuzzy logic for the analysis.

13. References

1. Dr. Sharvari Deshmukh
Assistant Professor, NIT Nagpur
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2. Dr. Raghavendra Deshmukh
Professor, VNIT Nagpur
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3. Dr. Nikhil Agrawal
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